

Practical No.6

MOHAMMAD QASIM

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Objective:

Write a program for checking whether the roots of quadratic equation are real or imaginary based on the given coefficients.

Python Code

```
In [14]: def Roots(a,b,c):  
          D=b**2-4*a*c  
          r1=(-b-D**0.5)/(2*a)  
          r2=(-b+D**0.5)/(2*a)  
          if D==0:  
              print("Roots are real and requal",r1,r2)  
          elif D>0:  
              print("Roots are real and distinct",r1,r2)  
          else:  
              print("Roots are imaginary or complex",r1,r2)
```

```
In [15]: Roots(1,5,-10)
```

Roots are real and distinct -6.531128874149275 1.5311288741492746

Interpretation:

As we can see the the defined function 'Roots' is giving roots of the quadratic equation as per its coefficient values.